A Positive Recurrence for the Fill–Janson Polynomial
A Complete Formal Verification of MathDB 372794

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AI-assisted verification note

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Abstract

We prove that every coefficient of the two-variable polynomial in Conjecture E.9 of Fill and Janson is nonnegative for every natural exponent. A homogeneous reformulation admits a triangular recurrence whose multipliers and initial values all have nonnegative coefficients. This gives a division-free induction and a complete Lean certificate, including the identity with the original polynomial. This is an independent verification of a known posted solution, not a new-discovery claim: Shivam Patel previously posted a positive generating-function proof on MathDB, and a public dossier by Machina Mathematica already reports a Lean proof of the same conjecture.

1 The exact question

Fill and Janson introduce, in Lemma E.8(a), equation (E.16) of [1],

$$h(t,a;r) = [1+(3+a)t]^r\big([1+(2+a)t]^{r+1} - [1+at]^{r+1}\big) \\ - [1+(1+a)t]^r\big([1+(4+a)t]^{r+1} - [1+(2+a)t]^{r+1}\big).$$

They prove pointwise nonnegativity for real $a, t \geq 0$ and real $r \geq 1$. Their Conjecture E.9 asks for coefficientwise nonnegativity when r is a natural number. These are different assertions: pointwise nonnegativity on a quadrant alone does not imply nonnegative coefficients. The conjecture is recorded as MathDB 372794 [2].

Theorem 1. *For every $r \in \mathbb{N}$ and every $i, j \in \mathbb{N}$,*

$$[t^i a^j]h(t,a;r) \geq 0.$$

In particular, the assertion includes $r = 0$, if that exponent is included in the convention for $\mathbb{N}$.

The proof is an identity in $\mathbb{Z}[t,a]$, so the coefficient domain and all index quantifiers are explicit. There is no hypothesis that a or t are numerical nonnegative inputs: here they are indeterminates.

2 A homogeneous sequence

Work first in $R = \mathbb{Z}[x,t]$. Set

$$A = (x+3t)(x+2t), \qquad B = (x+t)(x+4t),$$
$$C = (x+t)(x+2t), \qquad D = x(x+3t),$$
$$K = 8t^4(x+2t).$$

Define

$$H_n = (x+2t)A^n - (x+4t)B^n + (x+2t)C^n - xD^n.$$

Although this expression has subtraction, the following recurrence proves that its coefficients are nonnegative. Define two auxiliary sequences by

$$U_n = H_{n+1} - DH_n, \qquad V_n = U_{n+1} - CU_n.$$

Lemma 2. *The initial values and recurrences are*

$$\begin{aligned} H_0 &= 0, & U_0 &= 0, & V_0 &= K, \\ H_{n+1} &= DH_n + U_n, \\ U_{n+1} &= CU_n + V_n, \\ V_{n+1} &= BV_n + KA^{n+1} \qquad (n \geq 0). \end{aligned}$$

Proof. Direct expansion gives $H_0 = H_1 = 0$ and $H_2 = K$, proving the initial assertions. The first two recurrences are rearrangements of the definitions. To prove the third, let E be the shift operator on sequences. On an exponential sequence Z^n, the operator $E - W$ acts by multiplication by $Z - W$. Consequently

$$(E-B)(E-C)(E-D)H_n = (x+2t)(A-B)(A-C)(A-D)A^n,$$

because the B^n, C^n, D^n terms vanish. The three differences are

$$A - B = 2t^2, \quad A - C = 2t(x+2t), \quad A - D = 2t(x+3t).$$

Their product, multiplied by $x + 2t$, is KA. Thus $V_{n+1} - BV_n = KA^{n+1}$, as required. The Lean proof verifies the same equality by ring normalization after expanding powers at successor indices; it assumes no theory of shift operators. □

3 Positivity and the source substitution

Let R_+ denote the polynomials in $\mathbb{Z}[x,t]$ whose coefficients are nonnegative. It contains zero, every natural scalar, x, and t. It is closed under addition and multiplication: each coefficient of a product is a finite sum of products of nonnegative integers. It is therefore also closed under natural powers.

All five polynomials A, B, C, D, K belong to R_+. Lemma 2 now gives simultaneously, by induction on n,

$$H_n, U_n, V_n \in R_+.$$

At zero this follows from the displayed initial values. At each successor it follows by addition and multiplication in the three recurrence equations. This proves coefficientwise positivity of H_n without any analytic argument or cancellation estimate.

To recover the original polynomial, substitute $x = 1 + at$. Each image of an indeterminate has nonnegative coefficients, so the substitution preserves coefficientwise nonnegativity. More explicitly, writing

$$L_j = 1 + (a+j)t \quad (j = 0, 1, 2, 3, 4),$$

one obtains $A = L_3L_2$, $B = L_1L_4$, $C = L_1L_2$, and $D = L_0L_3$. Hence

$$\begin{aligned} H_r\big|_{x=1+at} &= L_2(L_3L_2)^r - L_4(L_1L_4)^r + L_2(L_1L_2)^r - L_0(L_0L_3)^r \\ &= L_3^r(L_2^{r+1} - L_0^{r+1}) - L_1^r(L_4^{r+1} - L_2^{r+1}) \\ &= h(t,a;r). \end{aligned}$$

This proves Theorem 1.

4 Relation to the earlier solution

Patel's MathDB response, posted on August 20, 2026, gives a generating-function proof. In the homogeneous notation above, its algebraic core is

$$\sum_{n\geq 0} H_n z^n = \frac{Kz^2}{(1-Az)(1-Bz)(1-Cz)(1-Dz)}.$$

The numerator and the four geometric-series factors have nonnegative coefficients. Our triangular recurrence is a finite inductive presentation of this same positive structure. We credit the posted solution as prior mathematical work, while the code in this packet was written locally using the recurrence rather than importing that response as an assumption.

A targeted literature search also located the public dossier [3], which states the full coefficient conjecture and reports prior Lean verification via a more general swap identity. Accordingly this packet makes no claim to the first proof, the first formalization, or a new theorem in the literature. The outcome is a complete, independently reproducible local formal verification of one distinct MathDB question.

5 Formal statement and proof coverage

The module `formal/CoefficientPositivity.lean` uses

$$\texttt{Poly = MvPolynomial (Fin 2) } \mathbb{Z}.$$

Coordinate 0 is t and coordinate 1 is a. The predicate `NonnegCoeffs p` means that the coefficient of every finitely supported exponent vector is a nonnegative integer. Every pair of exponents (i, j) determines exactly such a vector; there are no other coordinates.

The helper declarations prove closure under addition, multiplication, constants, variables, and powers. The theorem `homogeneous_nonnegative` is stronger than needed: it permits any two polynomials x, t with nonnegative coefficients in the same bivariate ring. Thus the formal proof uses the already-substituted $x = 1 + at$ directly. It does not need to assume a separate substitution-positivity lemma. The declaration `source_bridge` proves equality with the precise four-factor expression in (E.16). The final theorem is

$$\forall (r : \mathbb{N})(d : \mathrm{Fin}(2) \to_0 \mathbb{N}), \qquad 0 \leq \mathrm{coeff}(\mathrm{sourcePolynomial}(r), d).$$

Here $\to_0$ denotes the finitely supported function type. This theorem quantifies over every exponent and every coefficient, including the constant coefficient and exponents outside the polynomial's support. No finite test range is substituted for the universal claim.

The pinned Lean 4.34.1 compiler checked all 24 local declarations, with warnings treated as errors. The dependency revisions and proof hash are recorded in `pins.json` and the release receipts. Only the standard logical foundations of mathlib (propositional extensionality, classical choice, and quotient soundness) occur in the axiom reports. No unproved source conjecture is imported. An independent review of the source interpretation and this manuscript accompanies the release; compilation alone is not such a review.

References

[1] J. A. Fill and S. Janson, *Positive autocorrelation at unit lag for stationary random walk Metropolis–Hastings in* $\mathbb{R}^d$, arXiv:2601.19323v1 (2026), Lemma E.8(a), equation (E.16), Conjecture E.9. https://arxiv.org/abs/2601.19323v1.

[2] MathDB, post 372794, *The nonnegative-coefficient conjecture for the polynomial* $h(t, a; r)$; solution by Shivam Patel, August 20, 2026. https://mathdb.com/p/372794.

[3] Machina Mathematica, released by Korea Superintelligence Labs, *Swap differences of powers: a conjecture of Fill and Janson on nonnegative coefficients, and two cases of their root conjecture*, version 1, source snapshot September 7, 2026. https://ideosphere.ai/papers/fill-janson-coeff.